Fe-C Problem 8 | Problems T4_PhaseDiagrams II.pdf #8 | Fe-C · alloying elements

Annealed 0.55 % C Steel Bar and the Effect of Nickel

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

8.- Examine closely the phase diagram Fe-C and answer the following questions. a) If there is a 5 kg annealed bar of an alloy with 0.55 % C, what is the quantity (mass expressed in kg) of pre-eutectoid ferrite, eutectoid ferrite and cementite that is contained at 726 ºC? b) List the phases and microconstituents in this steel, their composition and their percentage at 600 ºC. c) How are the carbon content and temperature at the eutectoid point modified in the diagram if Ni is added? How does this addition modify the steel's microstructure?

Official Solution: a) α' = 1.43 kg; pearlite = 3.6 kg; α_{total} = 4.6 kg; $\alpha_{eutectoid}$ = 3.17 kg; Fe_3C = 0.4 kg; b) $x_{\alpha pre}$ = 0.26; $x_{eutectoid}$ = 0.72; $w_{\alpha total}$ = 0.92; w_{Fe_3C} = 0.08; c) possible decrease in amount of primary ferrite

PHASE 1 · HYPOTHESES & PARAMETERS

1. **Annealed** means slowly cooled: equilibrium structure, Fe-Fe3C diagram [Session 9 Slide 4].
2. $C_0 = 0.55 < 0.76$: hypoeutectoid; at 726 °C (just below 727 °C) the structure is proeutectoid ferrite + pearlite [Session 9 Slide 14].
3. At 600 °C the ferrite solvus gives slightly less than 0.022 % C; the difference does not change the result to two decimals, so 0.022 % C is used (a reading of the figure).
4. Data: m = 5 kg.

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C); tie lines just above 727 °C (α 0.022 / γ 0.76, for the proeutectoid ferrite) and below it (α 0.022 / Fe3C 6.67, for the phases). Masses are fractions times 5 kg.

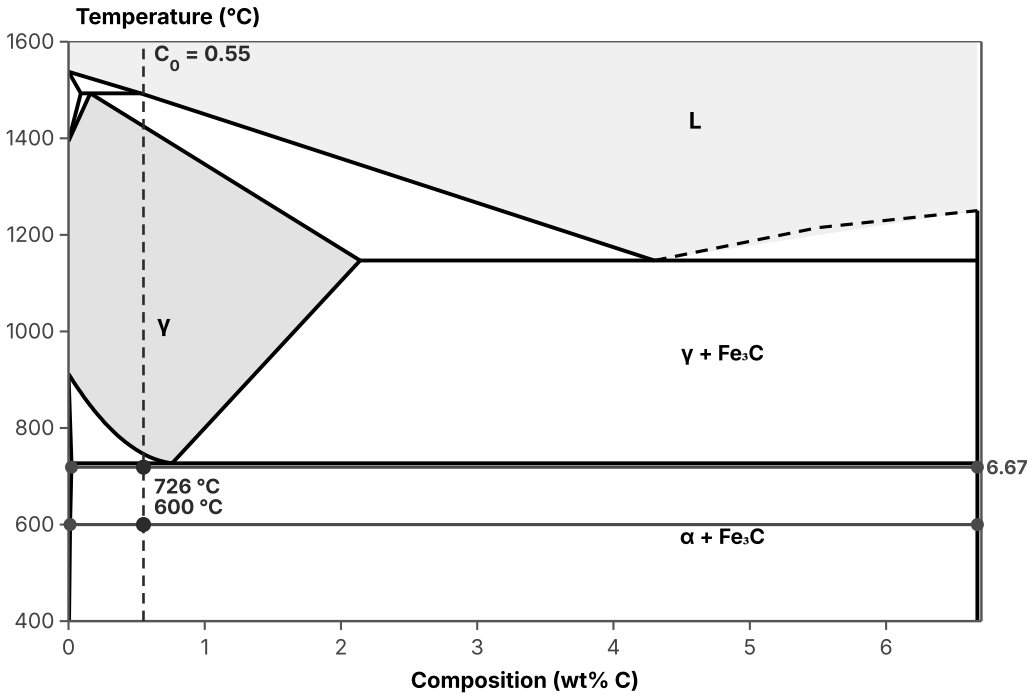


Figure 36. Fe-0.55 % C at 726 °C and at 600 °C: α + Fe_3C, as proeutectoid ferrite and pearlite.

PHASE 3 · STEP-BY-STEP DERIVATION

a) At 726 °C

$$W_{\alpha'} = \frac{0.76 - 0.55}{0.76 - 0.022} = \frac{0.21}{0.738} = 0.285 \;\Rightarrow\; m_{\alpha'} = 0.285 \times 5 = 1.42 \text{ kg}$$

$$W_{\text{pearlite}} = 1 - 0.285 = 0.715 \;\Rightarrow\; m_{\text{pearlite}} = 0.715 \times 5 = 3.575 \approx 3.58 \text{ kg}$$

$$W_{\alpha,\text{total}} = \frac{6.67 - 0.55}{6.648} = \frac{6.12}{6.648} = 0.921 \;\Rightarrow\; m_{\alpha,\text{total}} = 4.60 \text{ kg}$$

$$m_{\alpha,E} = 4.60 - 1.42 = 3.18 \text{ kg} \qquad m_{\mathrm{Fe_3C}} = (1 - 0.921) \times 5 = 0.079 \times 5 = 0.40 \text{ kg}$$

b) At 600 °C

- **Phases:** α (≈ 0.022 % C) 92 % and Fe3C (6.67 % C) 8 %.
- **Microconstituents:** proeutectoid ferrite 28.5 % and pearlite (0.76 % C overall) 71.5 %. The microconstituents formed at 727 °C do not change on further slow cooling.

c) Effect of nickel Ni is a **γ-stabilising element**: it displaces the eutectoid point to **lower temperature and lower %C** [Session 9 Slides 22-23]. With a lower eutectoid composition, the same 0.55 % C steel lies closer to the eutectoid point, so the short tie line gives **less proeutectoid ferrite and more pearlite**.

PHASE 4 · CHECKS & INTERPRETATION

- Mass check: $1.42 + 3.58 = 5.00$ kg; $4.60 + 0.40 = 5.00$ kg; pearlite $3.18 + 0.40 = 3.58$ kg .
- Carbon, with the unrounded masses $m_{\mathrm{Fe_3C}} = 0.0794 \times 5 = 0.397$ kg and $m_\alpha = 4.603$ kg: $0.397 \times 0.0667 + 4.603 \times 0.00022 = 0.02649 + 0.00101 = 0.0275$ kg C $= 0.55\,\%$ of 5 kg .
- **Limiting cases** (fractions dimensionless, masses in kg, compositions in wt% C): at $C_0 = 0.022$ the steel would be all proeutectoid α and at $C_0 = 0.76$ all pearlite; 0.55 % C lies nearer the eutectoid end, hence 0.715 of pearlite.

Check against the official solution

Part (a) agrees with the key in 3.6, 4.6 and 0.4 kg; (c) agrees.

DISCREPANCY WITH THE OFFICIAL SOLUTION

The key gives 1.43 kg of proeutectoid ferrite and 3.17 kg of eutectoid ferrite. Here $0.2846 \times 5 = 1.423 \to 1.42$ kg and $4.60 - 1.42 = 3.18$ kg; the key's 3.17 follows from its 1.43. The gap is 0.01 kg in the last digit; the computed values are kept.

DISCREPANCY WITH THE OFFICIAL SOLUTION

For part (b) the key gives $x_{\alpha pre} = 0.26$ and $x_{eutectoid} = 0.72$, which do not add up to 1 and do not match its own part (a) ($1.43/5 = 0.286$, $3.6/5 = 0.72$). The lever rule gives 0.285 proeutectoid ferrite and 0.715 pearlite, the same at 726 and at 600 °C; these values are kept.